

Functions & Graphs

ANSWER KEY



Evaluating functions

- 1 If $f(x) = 3x^2 - 2x + 5$, find $f(-2)$.
 $f(-2) = 3(4) - 2(-2) + 5 = 12 + 4 + 5 = 21$.
- 2 If $g(x) = \frac{2x+1}{5-x}$, find $g(5)$.
 $g(5) = \frac{2(5)+1}{5-3} = \frac{11}{2}$.

Domain and range

- 3 Find the domain of $f(x) = \frac{4}{x+7}$.
The denominator cannot equal zero: $x + 7 \neq 0$, so $x \neq -7$. Domain: all reals except $x = -7$.
- 4 Find the domain of $g(x) = \sqrt{2x-8}$.
The radicand must be non-negative: $2x - 8 \geq 0$, so $x \geq 4$. Domain: $x \geq 4$.

Composite functions

- 5 If $f(x) = 2x + 1$ and $g(x) = x^2 - 3$, find $f(g(2))$.
First, $g(2) = 4 - 3 = 1$. Then $f(1) = 2(1) + 1 = 3$.
- 6 If $f(x) = x + 4$ and $g(x) = 3x$, find $(f \circ g)(x)$ and $(g \circ f)(x)$, and explain why they differ.
 $(f \circ g)(x) = f(3x) = 3x + 4$. $(g \circ f)(x) = g(x + 4) = 3(x + 4) = 3x + 12$. They differ because function composition is generally not commutative — the order in which functions are applied changes the result.

Graphing linear and quadratic functions

- 7 Find the x - and y -intercepts of $y = 2x - 6$.
 y -intercept (set $x = 0$): $y = -6$, so $(0, -6)$. x -intercept (set $y = 0$): $0 = 2x - 6$, so $x = 3$, giving $(3, 0)$.
- 8 Find the vertex of $y = x^2 - 8x + 15$.
The vertex's x -coordinate is $-\frac{b}{2a} = -\frac{-8}{2} = 4$. $y = 16 - 32 + 15 = -1$. Vertex: $(4, -1)$.

Transformations of graphs

- 9 The graph of $y = x^2$ is shifted 4 units left and 3 units down. Write the equation of the resulting graph.
Shifting left by 4 replaces x with $x + 4$; shifting down 3 subtracts 3: $y = (x + 4)^2 - 3$.

- 10 Describe how the graph of $y = -3(x - 2)^2 + 1$ differs from the graph of $y = x^2$.

The graph is reflected over the x -axis and vertically stretched by a factor of 3 (from the -3), shifted 2 units right (from $x - 2$), and shifted 1 unit up (from $+1$).

Piecewise and step functions

- 11 A function is defined as $f(x) = 2x$ for $x \leq 2$ and $f(x) = x + 5$ for $x > 2$. Find $f(0)$ and $f(4)$.

Since $0 \leq 2$, use the first piece: $f(0) = 2(0) = 0$. Since $4 > 2$, use the second piece: $f(4) = 4 + 5 = 9$.

Interpreting graphs in context

- 12 A graph shows a car's speed over time: the speed increases from 0 to 40 mph over the first 10 seconds, stays constant at 40 mph for the next 20 seconds, then decreases to 0 over the final 10 seconds. Describe what is happening to the car during each phase.

During the first phase (0-10 seconds), the car is accelerating from a stop up to 40 mph. During the second phase (10-30 seconds), the car travels at a constant speed of 40 mph (cruising). During the third phase (30-40 seconds), the car is decelerating, coming to a stop.

Finding the equation of a function from a graph description

- 13 A parabola has x -intercepts at $x = -1$ and $x = 5$, and passes through $(2, -9)$. Find its equation in factored form.

Factored form: $y = a(x + 1)(x - 5)$. Substitute $(2, -9)$: $-9 = a(3)(-3) = -9a$, so $a = 1$. Equation: $y = (x + 1)(x - 5)$.

Multi-part function analysis

- 14 This question has two parts. A function is given by $f(x) = x^2 - 4x - 5$. (a) Find the zeros of the function by factoring. (b) Find the vertex, and state whether it is a minimum or maximum.

(a) Factor: $x^2 - 4x - 5 = (x - 5)(x + 1) = 0$, giving zeros at $x = 5$ and $x = -1$. (b) Vertex x -coordinate: $-\frac{-4}{2} = 2$. $f(2) = 4 - 8 - 5 = -9$. Vertex: $(2, -9)$, which is a minimum since the parabola opens upward (positive leading coefficient).

Inverse functions

- 15 Find the inverse of $f(x) = 4x + 8$.

Let $y = 4x + 8$. Swap x and y : $x = 4y + 8$. Solve for y : $x - 8 = 4y$, so $y = \frac{x - 8}{4}$. Thus $f^{-1}(x) = \frac{x - 8}{4}$.

- 16 If $f(3) = 10$, find $f^{-1}(10)$, using the definition of an inverse function.
By definition, $f^{-1}(10) = 3$, since inverse functions reverse input-output pairs.
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Rational and radical functions

- 17 Find the vertical asymptote of $f(x) = \frac{3}{x-5}$.
The vertical asymptote occurs where the denominator equals zero: $x - 5 = 0$, so $x = 5$.
- 18 Find $g(9)$ for $g(x) = \sqrt{x} + 4$.
 $g(9) = \sqrt{9} + 4 = 3 + 4 = 7$.
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Exponential functions

- 19 If $f(x) = 2^x$, find $f(5)$ and $f(-2)$.
 $f(5) = 2^5 = 32$. $f(-2) = 2^{-2} = \frac{1}{4}$.
- 20 A function is given by $g(x) = 100(0.5)^x$. Find $g(3)$ and interpret its meaning if this models a decaying quantity over x time periods.
 $g(3) = 100(0.5)^3 = 100(0.125) = 12.5$. This means after 3 time periods, the original quantity of 100 has decayed to 12.5 — consistent with the quantity halving each period.
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Multi-part analysis of a real-world function model

- 21 This question has two parts. A ball is thrown, and its height is modeled by $h(t) = -16t^2 + 48t + 4$, where t is time in seconds. (a) Find the height of the ball at $t = 0$ and interpret this value. (b) Find the time at which the ball reaches its maximum height, and find that maximum height.
- (a) $h(0) = 4$, meaning the ball starts at a height of 4 feet (perhaps thrown from a platform or by someone's hand). (b) Maximum occurs at $t = -\frac{48}{2(-16)} = 1.5$ seconds.
 $h(1.5) = -16(2.25) + 48(1.5) + 4 = -36 + 72 + 4 = 40$ feet.